Quantitative remainders for small-disorder Liouville first passage percolation

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Abstract

Let $\lambda_d(\zeta)$ be the distance exponent of Liouville first passage percolation in dimension $d \geq 2$, with logarithmic variance coefficient one, and let $k_d = \lim_{\zeta \downarrow 0} \zeta^{-4/3} \lambda_d(\zeta)$. Combining the dyadic Gaussian variational problem of [A] with the finite-layer action and multiscale comparisons of [B], we prove

$$-C_d \zeta^{13/9} [\log(1/\zeta)]^{1/3} \leq \lambda_d(\zeta) - k_d \zeta^{4/3} \leq C_d \zeta^{5/3} [\log(1/\zeta)]^{1/2}.$$

The first new input bounds the error in each variational approximation of k_d by $O_d(\log N/N)$. The second bounds the integrated fourth power of the transverse slope of a geodesic in a weight of small oscillation. It improves the upper remainder in [B] by a factor of $\log(1/\zeta)$. All new estimates and their dependence on the block parameters are proved below; the precise inputs from the two manuscripts are stated explicitly.

1 Statement and inputs

Fix $d \geq 2$, and put $m = d - 1$. Constants denoted by C_d, c_d are positive, depend only on d, and may change between occurrences. All logarithms without a subscript are natural.

For definiteness, use the heat-kernel regularization. If W is white noise on $\mathbb{R}^d \times (0, \infty)$ with Lebesgue control measure, set

$$h_\varepsilon(x) = \int_\varepsilon^1 \int_{\mathbb{R}^d} t^{-(d+1)/2} k_H((x-y)/t)\, W(dy, dt), \qquad k_H(x) = (2/\pi)^{d/4} e^{-|x|^2}. \tag{1.1}$$

Then $\operatorname{Var} h_\varepsilon(x) = \log(1/\varepsilon)$. Write $\Omega = (-2, 2)^d$ and let D_ε^ζ be the internal distance from 0 to e_1 in Ω for the weight $e^{\zeta h_\varepsilon}$, with Euclidean arclength as reference measure and no Wick normalization. The exponent is characterized by

$$\frac{\log D_\varepsilon^\zeta}{\log \varepsilon} \longrightarrow \lambda_d(\zeta) \quad \text{in probability as } \varepsilon \downarrow 0.$$

Its fixed-parameter existence and its invariance under the regularizations used in the two manuscripts are inputs; see [A, Appendix A] and [B, Section 2]. In particular, this is the exponent denoted by $\chi_d(\xi)$ in [B], with $\xi = \zeta$.

Theorem 1.1. *For each $d \geq 2$ there are $C_d < \infty$ and $\zeta_0(d) > 0$ such that, for $0 < \zeta < \zeta_0(d)$, with $\ell = \log(1/\zeta)$,*

$$\boxed{-C_d \zeta^{13/9} \ell^{1/3} \leq \lambda_d(\zeta) - k_d \zeta^{4/3} \leq C_d \zeta^{5/3} \ell^{1/2}.} \tag{1.2}$$

Here $k_d \in (0, \infty)$ is the common leading constant identified in [A, Theorem 1.1] and [B, Theorem 1.1].

We specify the variational inputs and the one existing quantitative LFPP lower comparison now. The geometric and probabilistic inputs for the upper comparison are stated where used in Section 5.

The rough dyadic problem. Let $V_{n,d}$ consist of the continuous functions $f : [0,1] \to \mathbb{R}^m$ which vanish at both endpoints and are affine on every dyadic interval of length 2^{-n}. Set $\mathcal{E}(f) = \frac{1}{2}\int_0^1 |f'|^2$. Let $\mathcal{G}_d$ be the centered Gaussian process anchored by $\mathcal{G}_d(0) = 0$ whose increment variance is

$$\mathbb{E}(\mathcal{G}_d(f) - \mathcal{G}_d(g))^2 = 2\pi \int_0^1 |f - g|. \tag{1.3}$$

Write $A_{n,d} = \mathbb{E}\sup_{f\in V_{n,d}}(\mathcal{G}_d(f) - \mathcal{E}(f))$. By [A, Lemma 2.1 and Theorem 1.1],

$$A_{n+r,d} \le A_{n,d} + A_{r,d}, \qquad k_d = \inf_{n\ge 1} \frac{A_{n,d}}{n\log 2}, \qquad k_d n \log 2 \le A_{n,d} \le C_d n. \tag{1.4}$$

The stationary finite-layer problem. For $0 < \varepsilon < 1$ define the spatial covariance on $\mathbb{R}^m$

$$C_\varepsilon(z) = \sqrt{\frac{\pi}{2}} \int_{\varepsilon^2}^1 u^{-1/2} e^{-|z|^2/(2u)}\, du. \tag{1.5}$$

Let b_ε be white in time with covariance $C_\varepsilon(x-y)\delta(t-s)$. For $a > 0$, its pinned minimal action of duration T is

$$\mathscr{A}_{\varepsilon,a}(T) = \inf_{X\in H_0^1((0,T);\mathbb{R}^m)} \left\{ \frac{a}{2}\int_0^T |X'|^2\, dt + \int_0^T b_\varepsilon(X(t),t)\, dt \right\}.$$

As in [B, (3.1)], the noise along a path is defined using the Brownian primitive in time. Supremums and infimums can equivalently be taken over countable dense families. Write $K_a(C_\varepsilon)$ for minus the limiting mean action per unit time and $K(C_\varepsilon) = K_1(C_\varepsilon)$. Proposition 3.2 and equation (3.8) of [B] give

$$K_a(C_\varepsilon) = \sup_{T>0} \frac{-\mathbb{E}\mathscr{A}_{\varepsilon,a}(T)}{T} = a^{-1/3} K(C_\varepsilon). \tag{1.6}$$

In particular, the supremum formula applies to arbitrarily small T. Put $K_{N,d} = K(C_{3^{-N}})$; this is $K_N^{(d-1)}$ in [B]. Their Theorem 4.1 and the identification of the leading constant give

$$0 \le K_{N,d} \le k_d N \log 3 + C_d \log(N+2), \qquad \lim_{N\to\infty} \frac{K_{N,d}}{N} = k_d \log 3. \tag{1.7}$$

The factor $\log 3$ converts the rate per layer to the rate per unit of logarithmic variance. The equality of the k_d in (1.4) and (1.7) uses the LFPP limit already established by the two manuscripts.

The existing LFPP lower comparison. Proposition 7.1 of [B], with the parameters in their equation (5.2), gives integers H_+, N_+ such that

$$\begin{aligned} H_+ &= \lfloor \zeta^{-4/9}\ell^{2/3} \rfloor, \qquad 0 \le H_+\log 2 - N_+ \log 3 \le C_d \ell, \\ \frac{\lambda_d(\zeta)}{\zeta^{4/3}} &\ge \frac{K_{N_+,d}}{H_+\log 2} - C_d \zeta^{1/9}\ell^{1/3}. \end{aligned} \tag{1.8}$$

The new lower bound will follow from a quantitative reverse inequality in (1.7).

2 Quantitative comparison of the variational problems

In this section write $A_n = A_{n,d}$ and $K_N = K_{N,d}$. Let $V_n(T)$ be the space of pinned, piecewise affine paths on $[0,T]$ with mesh $h = T2^{-n}$, and let $\mathcal{E}_T(X) = \frac{1}{2}\int_0^T |X'|^2$.

2.1 The exact Gaussian decomposition

Fix $\varepsilon = 3^{-N}$ and set

$$q_N(z) = 2[C_\varepsilon(0) - C_\varepsilon(z)].$$

Define $q_-(z)$ and $q_+(z)$ by integrating the expression

$$\sqrt{2\pi}\, u^{-1/2}(1 - e^{-|z|^2/(2u)})$$

over $(0, \varepsilon^2)$ and $(1, \infty)$, respectively. Direct integration and $1 - e^{-v} \le \min(1, v)$ give

$$q_-(z) + q_N(z) + q_+(z) = 2\pi|z|, \quad q_-(z) \le C\min(\varepsilon, |z|), \quad q_+(z) \le C|z|^2. \tag{2.1}$$

For the identity, substituting $v = |z|^2/(2u)$ reduces the integral to $\sqrt{\pi}\,|z| \int_0^\infty v^{-3/2}(1 - e^{-v})\, dv = 2\pi|z|$.

Each q_i is a Gaussian increment variance. For q_+, the unanchored point variance is infinite; anchoring the spatial field at zero before integrating gives a finite covariance $[q_+(x) + q_+(y) - q_+(x-y)]/2$. Consequently, there are independent centered Gaussian path processes G_-, G_N, G_+, each zero at the zero path, with

$$\mathbb{E}(G_i(X) - G_i(Y))^2 = \int_0^T q_i(X(t) - Y(t))\, dt. \tag{2.2}$$

Their sum G has increment variance $2\pi\int_0^T |X - Y|$. In the finite-layer component we may take

$$G_N(X) = -\int_0^T [b_\varepsilon(X(t), t) - b_\varepsilon(0, t)]\, dt.$$

Under $X(t) = Tf(t/T)$, both the rough Gaussian process and the kinetic energy scale by T, so

$$\mathbb{E} \sup_{X \in V_n(T)} \{G(X) - \mathcal{E}_T(X)\} = TA_n. \tag{2.3}$$

All processes here have separable versions continuous on bounded coefficient sets. Gaussian comparisons below can first be made on finite coefficient grids with the common deterministic penalty, and then passed to their increasing union. The value at zero is zero, so the resulting supremums are nonnegative.

2.2 Bounds for the two tails

Lemma 2.1. *Let O be a centered stationary Gaussian process on $\mathbb{R}$ with covariance $\mathbb{E}O(x)O(y) = e^{-|x-y|}$. For every $\beta > 0$,*

$$\mathbb{E}\sup_{x \in \mathbb{R}}\{O(x) - O(0) - \beta x^2\} \le C\sqrt{\log(2 + \beta^{-1})}. \tag{2.4}$$

Proof. Let $S_j = \sup_{[j,j+1]} O$. Stationarity gives $\mathbb{E}S_j = \mu < \infty$, and Gaussian concentration gives $\mathbb{E}e^{vS_j} \le e^{v\mu+v^2/2}$ for $v > 0$. Subtracting $O(0)$ does not affect the expected supremum. Taking a logarithmic moment and summing over integer intervals yields

$$\begin{aligned}\mathbb{E}\sup_x\{O(x) - \beta x^2\} &\le \mu + \frac{v}{2} + \frac{1}{v}\log\sum_{j\in\mathbb{Z}} e^{-v\beta((|j|-1)_+)^2} \\ &\le \mu + \frac{v}{2} + \frac{1}{v}\log[C(1+(v\beta)^{-1/2})].\end{aligned}$$

Choose $v = \sqrt{\log(2+\beta^{-1})}$. No independence between the S_j is used. □

Lemma 2.2. *If $h = T2^{-n} \ge \varepsilon$ and $0 < \eta \le 1$, then*

$$\frac{1}{T}\mathbb{E}\sup_{X\in V_n(T)}\{G_-(X) - \eta\mathcal{E}_T(X)\} \le C_d\sqrt{\frac{\varepsilon}{h}}\sqrt{\log(2+\eta^{-1}(h/\varepsilon)^{3/2})}. \tag{2.5}$$

For every $T, \eta > 0$,

$$\mathbb{E}\sup_{X\in V_n(T)}\{G_+(X) - \eta\mathcal{E}_T(X)\} \le C_d T^2/\eta. \tag{2.6}$$

Proof. Expand each coordinate of X in height-one Schauder tents $\psi_{j,k}$ of support length $\ell_j = T2^{-j}$, $0 \le j < n$: $X_i = \sum_{j,k} a_{i,j,k}\psi_{j,k}$. Orthogonality of their derivatives gives

$$\mathcal{E}_T(X) = 2\sum_{i,j,k}\frac{a_{i,j,k}^2}{\ell_j}. \tag{2.7}$$

The first bound in (2.1), the subadditivity of $x \mapsto \min(\varepsilon, x)$ on $[0,\infty)$, and $0 \le \psi_{j,k} \le 1$ imply

$$\int_0^T q_-(X-Y)\,dt \le C\sum_{i,j,k}\ell_j\min(\varepsilon, |a_{i,j,k} - \widetilde{a}_{i,j,k}|).$$

This metric is dominated by that of the sum of independent processes

$$c\sqrt{\varepsilon\ell_j}\,[O_{i,j,k}(a_{i,j,k}/\varepsilon) - O_{i,j,k}(0)],$$

where c is a sufficiently large absolute constant, since $1 - e^{-r} \ge c_0\min(1,r)$ for $r \ge 0$. Sudakov–Fernique, including the common deterministic penalty $-\eta\mathcal{E}_T$, and (2.7) reduce the supremum to the sum of one-coefficient supremums. In each, put $a = \varepsilon x$. After dividing by $c\sqrt{\varepsilon\ell_j}$, the quadratic coefficient is $\beta = (2/c)\eta(\varepsilon/\ell_j)^{3/2}$. Lemma 2.1 therefore bounds the mean contribution by

$$C\sqrt{\varepsilon\ell_j}\sqrt{\log(2+\eta^{-1}(\ell_j/\varepsilon)^{3/2})}.$$

There are T/ℓ_j tents per coordinate at level j. Write $\ell_j = 2^k h$, $1 \le k \le n$, divide by T, and use

$$\log(2 + a2^{3k/2}) \le \log(2+a) + (3k/2)\log 2.$$

The summability of $2^{-k/2}(1+\sqrt{k})$ proves (2.5).

For (2.6), the last bound in (2.1) allows comparison with the linear Gaussian functional $\sqrt{C}\int_0^T X(t)\cdot dB(t)$, where B is an m-dimensional Brownian motion. Enlarge the path space

to $H_0^1((0,T);\mathbb{R}^m)$. In the L^2-normalized Dirichlet sine basis, the eigenvalues of $-d^2/dt^2$ are $\lambda_j = (\pi j/T)^2$. Maximizing each coefficient and taking expectations gives

$$\frac{Cm}{2\eta}\sum_{j\geq 1}\lambda_j^{-1} = \frac{CmT^2}{12\eta}.$$

The finite trace justifies the passage from finite sine sums to the whole space, and proves the claim. □

Proposition 2.3 (Uniform variational comparison)**.** *For $N, n \geq 1$, $T > 0$, $0 < \eta < 1/2$, and $h = T2^{-n} \geq \varepsilon = 3^{-N}$,*

$$\boxed{A_n \leq (1-2\eta)^{-1/3}K_N + C_d\left\{\frac{T}{\eta} + \sqrt{\frac{\varepsilon}{h}}\sqrt{\log\bigl(2+\eta^{-1}(h/\varepsilon)^{3/2}\bigr)}\right\}.} \tag{2.8}$$

Proof. Split $G = G_N + G_- + G_+$ and allocate the fractions $1-2\eta, \eta, \eta$ of the kinetic energy to these three terms. The supremum of the sum is at most the sum of the supremums. For the first, enlarge $V_n(T)$ to $H_0^1((0,T);\mathbb{R}^m)$. Its anchored value differs from $-\mathscr{A}_{\varepsilon,1-2\eta}(T)$ by the centered noise along the zero path. Hence (1.6) gives

$$\mathbb{E}\sup_{V_n(T)}\{G_N - (1-2\eta)\mathcal{E}_T\} \leq T(1-2\eta)^{-1/3}K_N.$$

Use Lemma 2.2 on the remaining terms and divide by T in (2.3). □

2.3 Rates for both approximations

Theorem 2.4. *For every $N, n \geq 1$,*

$$|K_{N,d} - k_d N\log 3| \leq C_d\log(N+2), \tag{2.9}$$

$$0 \leq A_{n,d} - k_d n\log 2 \leq C_d\log(n+2). \tag{2.10}$$

Proof. For sufficiently large N, set

$$T = \eta = (N+2)^{-1}, \qquad n = \left\lfloor \log_2\frac{3^N}{(N+2)^2}\right\rfloor. \tag{2.11}$$

Then $n \geq 1$ and $N+2 \leq h/\varepsilon < 2(N+2)$. The error in Proposition 2.3 is at most C_d, since $T/\eta = 1$ and the other term is $O(\sqrt{\log(N+2)/(N+2)})$. Also $K_N \leq C_d(N+1)$ by (1.7), so replacing the coefficient $(1-2\eta)^{-1/3}$ by one costs at most C_d. Using (1.4), we obtain

$$K_N \geq k_d n\log 2 - C_d \geq k_d N\log 3 - 2k_d\log(N+2) - C_d. \tag{2.12}$$

Together with (1.7), this proves (2.9). Increasing C_d handles the finitely many smaller N.

For (2.10), given sufficiently large n, take

$$N = \left\lceil \frac{n\log 2 + 3\log(n+2)}{\log 3}\right\rceil, \qquad T = \eta = (N+2)^{-1}.$$

Then $N \asymp n+2$ and $h/\varepsilon \asymp (n+2)^2$. The tail terms in Proposition 2.3 are bounded, and the kinetic coefficient again costs $O_d(1)$. Thus

$$A_n \leq K_N + C_d \leq k_d N\log 3 + C_d\log(N+2) \leq k_d n\log 2 + C_d\log(n+2).$$

The lower bound is (1.4); the remaining finitely many n are absorbed in the constant. □

3 The lower remainder

Apply (2.9) to the integers in (1.8). For sufficiently small ζ, $N_+ \asymp H_+$ and $\log(N_+ + 2) \le C_d \ell$. Therefore

$$\frac{K_{N_+,d}}{H_+ \log 2} \ge k_d - C_d \frac{\ell}{H_+} \ge k_d - C_d \zeta^{4/9} \ell^{1/3}.$$

Substitution in (1.8), followed by multiplication by $\zeta^{4/3}$, gives

$$\lambda_d(\zeta) \ge k_d \zeta^{4/3} - C_d \zeta^{13/9} \ell^{1/3}. \tag{3.1}$$

Here the finite-layer deficit contributes only $O_d(\zeta^{16/9} \ell^{1/3})$. The larger error in (3.1) comes from the existing local LFPP comparison.

4 An integrated estimate for geodesics

We first prove the interpolation inequality needed to improve the geometric step of [B, Lemma 5.3 and Proposition 5.5].

Lemma 4.1. *Let I be a bounded interval and $f : I \to \mathbb{R}^m$ a bounded measurable function. For subintervals J, write $f_J = |J|^{-1} \int_J f$, and set*

$$B^2 = \sup_{\substack{J \subset I \\ J \text{ dyadic}}} \frac{1}{|J|} \int_J |f - f_J|^2,$$

where the dyadic intervals include I itself. Then

$$\int_I |f|^4 \le 120 (B^2 + |f_I|^2) \int_I |f|^2. \tag{4.1}$$

Proof. Let $t_0 = |I|^{-1} \int_I |f|$ and let M_D be the dyadic maximal operator on I. For $t \ge t_0$, the maximal dyadic intervals J with $|J|^{-1} \int_J |f| > t$ have average at most $2t$, since their parents have average at most t. They cover $\{M_D |f| > t\}$ up to a null set, and $|f| \le t$ off their union. On each such J, $(|f| - 2t)_+ \le |f - f_J|$. Consequently

$$\int_I (|f| - 2t)_+^2 \le B^2 |\{M_D |f| > t\}|. \tag{4.2}$$

For $a \ge 0$, $\int_0^\infty t(a - 2t)_+^2 \, dt = a^4/48$. Multiply (4.2) by t and integrate over $[t_0, \infty)$. The omitted part of the left side is at most $(t_0^2/2) \int_I |f|^2$. It follows that

$$\frac{1}{48} \int_I |f|^4 \le \frac{t_0^2}{2} \int_I |f|^2 + \frac{B^2}{2} \int_I (M_D |f|)^2. \tag{4.3}$$

For completeness, $\|M_D g\|_2 \le 2 \|g\|_2$ for bounded $g \ge 0$: maximal intervals give $t |\{M_D g > t\}| \le \int_{\{M_D g > t\}} g$ for every $t > 0$, including when the whole interval is selected. Integration and Cauchy–Schwarz give

$$\|M_D g\|_2^2 \le 2 \int_I g M_D g \le 2 \|g\|_2 \|M_D g\|_2.$$

Finally $t_0^2 \le |I|^{-1} \int_I |f|^2 \le B^2 + |f_I|^2$. Using these facts in (4.3) proves (4.1). □

Proposition 4.2. *Let $Q = V \times [0, L]$, where $L > 0$ and $V \subset \mathbb{R}^m$ is a compact convex rectangle with nonempty interior. Let u be continuous on Q and $\|u\|_\infty \le M$, where $0 < M \le M_0(d)$. Let D be the crossing distance between the bottom and top faces of Q for the weight e^u, with free endpoints. There is $X \in H^1((0, L); V)$ such that*

$$D \ge L + \int_0^L u(X(z), z)\, dz + \frac{1 - C_d M}{2} \int_0^L |X'(z)|^2\, dz - C_d M e^{-c_d/\sqrt{M}} L. \tag{4.4}$$

The constant $M_0(d)$ can be chosen so that $1 - C_d M \ge 1/2$.

Proof. If u is constant, a vertical segment and $e^u \ge 1+u$ prove the claim. Otherwise put $\theta = \operatorname{osc}_Q u$, so $0 < \theta \le 2M$. A minimizing crossing exists by compactness and lower semicontinuity of weighted length. Parametrize one by Euclidean arclength $t \in [0, \mathfrak{l}]$, and write its unit tangent as $\sigma = (\sigma_x, \sigma_z)$. Every subarc minimizes between its endpoints. Since its chord stays in Q, comparison with the chord gives, for every interval $J \subset [0, \mathfrak{l}]$,

$$|\sigma_J| \ge e^{-\theta}, \qquad \frac{1}{|J|} \int_J |\sigma - \sigma_J|^2 = 1 - |\sigma_J|^2 \le C\theta. \tag{4.5}$$

Comparison with a vertical segment gives $\mathfrak{l} \le e^\theta L$. As $\int_0^{\mathfrak{l}} \sigma_z = L$, the full-interval mean has longitudinal component at least $e^{-\theta}$, and hence $|(\sigma_x)_{[0,\mathfrak{l}]}|^2 \le 1 - e^{-2\theta} \le C\theta$. Lemma 4.1 applied to σ_x yields

$$\int_0^{\mathfrak{l}} |\sigma_x|^4\, dt \le C\theta \int_0^{\mathfrak{l}} |\sigma_x|^2\, dt. \tag{4.6}$$

Use the chord replacement in the proof of [B, Lemma 5.3], with the fixed slope threshold $\alpha_0 = 1/2$. Its hypothesis $C_d\sqrt{\theta} \le \alpha_0$ holds after decreasing M_0. The intervals replaced form a relatively open set O with

$$|O| \le C_d e^{-c_d/\sqrt{\theta}} \mathfrak{l}. \tag{4.7}$$

Keep the original arclength parameter on the new curve. Its derivative τ equals σ off O and equals σ_J on every component J of O. The construction gives

$$c \le \tau_z \le 1, \qquad |\tau_x|/\tau_z \le 1/2 \quad \text{almost everywhere.}$$

Indeed, off O the longitudinal component is at least $3/4$. On each replaced interval the proportion of tangents with longitudinal component below $3/4$ or transverse component above $\alpha_0/4$ is at most $\alpha_0/16$. This bounds the longitudinal chord derivative away from zero. The entire-interval chord, if selected, has longitudinal derivative at least $e^{-\theta}$. The exponential bound (4.7) is the John–Nirenberg and maximal operator estimate in that lemma; no parameter tending to zero is used for α_0.

The new curve stays in Q by convexity. Its weighted length $\widetilde{D}$ satisfies

$$\widetilde{D} \le D + (\max_Q e^u - \min_Q e^u)|O| \le D + CM|O|. \tag{4.8}$$

Jensen's inequality and (4.5) give

$$\int_0^{\mathfrak{l}} |\tau_x|^4\, dt \le \int_0^{\mathfrak{l}} |\sigma_x|^4\, dt, \tag{4.9}$$

$$\int_0^{\mathfrak{l}} |\sigma_x|^2\, dt - \int_0^{\mathfrak{l}} |\tau_x|^2\, dt = \sum_{J \text{ component of } O} \int_J |\sigma_x - (\sigma_x)_J|^2\, dt \le C\theta|O|. \tag{4.10}$$

The longitudinal coordinate of the new curve is strictly increasing, so it is a graph $z \mapsto (X(z), z)$. Put $v(z) = X'(z) = \tau_x(t)/\tau_z(t)$ and use $dz = \tau_z(t)\,dt$. The uniform bounds on τ_z, followed by (4.6)–(4.10), imply

$$\int_0^L |v|^4\,dz \le C\theta \int_0^L |v|^2\,dz + C\theta^2|O|. \tag{4.11}$$

Here $\int |\tau_x|^2 dt \le \int |v|^2 dz$ because $\tau_z \le 1$. Thus the quadratic energy of the new graph, rather than that of the original curve, appears on the right side.

For $a \ge 0$, $\sqrt{1+a} \ge 1 + a/2 - a^2/8$. Consequently

$$e^u\sqrt{1+|v|^2} \ge 1 + u + \tfrac{1}{2}e^{-M}|v|^2 - \tfrac{1}{8}e^M|v|^4.$$

Integrate this inequality and use (4.11). The coefficient of $\frac{1}{2}\int |v|^2$ is at least $1 - C_dM$. Finally (4.8), $\theta \le 2M$, $\mathfrak{l} \le e^\theta L$, and (4.7) absorb the remaining terms into $C_dMe^{-c_d/\sqrt{M}}L$. □

5 The improved upper remainder

We redo the local crossing estimate of [B, Proposition 5.5] with a longer block. Proposition 4.2 is the only new geometric input. Throughout this section set

$$s = \zeta^{2/3}, \quad H = \lfloor \zeta^{-2/3}\ell^{1/2} \rfloor, \quad R = B\zeta^{1/3}\ell^{1/2}, \quad g = \zeta^{4/3}H\log 2. \tag{5.1}$$

The constant $B = B(d)$ is sufficiently large and fixed; it will be chosen to meet the cover conditions in Section 5.3. In particular, $R, g \to 0$ and

$$Rg = (1 + o(1))B\zeta\ell\log 2. \tag{5.2}$$

5.1 A uniform action bound and one-rectangle crossings

We recall the exact input [B, Lemma 5.4]. For $T, W > 0$, $0 < a \le 1$, $0 < \eta \le 1$, and either the white-in-time field of covariance C_ε or the rescaled heat field $b_s(x,t) = s^{-1/2}h_\varepsilon(x, t/s)$, let

$$J_a = \inf_{X \in H^1((0,T);[0,W]^m)} \left\{ \frac{a}{2}\int_0^T |X'|^2\,dt + \int_0^T b_s(X(t),t)\,dt \right\}.$$

Endpoints are free. That lemma gives, uniformly in these parameters,

$$\mathbb{E}J_a \ge -(1+2\eta)Ta^{-1/3}K(C_\varepsilon) - C_da\eta T - C_d\sqrt{T\log(2 + W/(\eta T))}, \tag{5.3}$$

$$\mathbb{P}[J_a < \mathbb{E}J_a - z\sqrt{T}] \le e^{-c_dz^2}, \qquad z \ge 0. \tag{5.4}$$

Spectral domination by the time-white field is part of that lemma; in particular, no limit in s or estimate on a selected path's slope is required to use (5.3). Also monotonicity in the cutoff and (1.7) give

$$K(C_{2^{-H}}) \le k_dH\log 2 + C_d\log(H+2). \tag{5.5}$$

Fix a translated and rotated rectangle with longitudinal length $L \in [1/2, 1]$ and positive transverse side lengths at most ten. Let D_{rect} be its longitudinal crossing distance for the weight $e^{\zeta G_H}$, where $G_H = h_{2^{-H}}$. The uniform Gaussian supremum estimate used in the proof of [B, Proposition 5.5] states

$$\mathbb{P}[\|G_H\|_\infty > C_dH] \le C_de^{-c_dH}. \tag{5.6}$$

On the complementary event, apply Proposition 4.2 with the deterministic bound $M = C_d\zeta H$. The remainder in that proposition is at most $\zeta^{10}L$ for small ζ, since $\zeta H \asymp \zeta^{1/3}\sqrt{\ell}$. Take the deterministic coefficient $a = 1 - C_d\zeta H \in [1/2, 1]$. Under $t = sz$, $T = sL$, the selected graph's noise and kinetic terms become s times its action for $b_s(x,t) = s^{-1/2}G_H(x, t/s)$, because $\zeta = s^{3/2}$. Enlarge to all H^1 paths in a transverse cube of side ten, with free endpoints. This yields the pathwise bound

$$D_{\mathrm{rect}} \geq L + sJ_a - \zeta^{10}L. \tag{5.7}$$

Only the selected graph in Proposition 4.2 needs the integrated slope estimate; taking this unrestricted infimum preserves the lower inequality.

Apply (5.3) with $\eta = R$, $W = 10$, and $\varepsilon = 2^{-H}$. Here $\log(2 + 10/(RT)) \leq C_d\ell$ and $T \asymp s$. The estimates needed to divide the mean errors by HT are

$$\begin{aligned} 1 - a &= O_d(\zeta H) = O_d(R), \\ \frac{\log(H+2)}{H} &= O(\zeta^{2/3}\ell^{1/2}) = o(R), \\ \frac{\sqrt{\ell}}{H\sqrt{s}} &= O(\zeta^{1/3}) = o(R), \qquad \frac{\eta}{H} = o(R). \end{aligned} \tag{5.8}$$

Equations (5.3), (5.5), and (5.8) therefore give

$$\mathbb{E}J_a \geq -HT(k_d \log 2 + C_d R). \tag{5.9}$$

For each fixed $Q > 0$, a crossing below $L[1 - (k_d + C(d,Q)R)g]$ forces, by (5.7) and (5.9), a downward deviation of J_a of size $D(d,Q)RHT$. The constant $D(d,Q)$ can be made arbitrarily large by increasing $C(d,Q)$. The confidence exponent is

$$R^2H^2T \asymp B^2\ell^2. \tag{5.10}$$

Thus (5.4) bounds this probability by $\frac{1}{2}e^{-(Q+2)\ell^2}$, after adjusting constants. The event in (5.6) has smaller probability because $H/\ell^2 \to \infty$, and $\zeta^{10} = o(Rg)$. We have proved, uniformly over translations and rotations,

$$\mathbb{P}[D_{\mathrm{rect}} < L\{1 - (k_d + C(d,Q)R)g\}] \leq e^{-(Q+2)\ell^2}. \tag{5.11}$$

5.2 Finite range and the simultaneous crossing event

We give the remaining parameter checks so that (5.11) can be used with the cover in [B, Proposition 6.1]. As in that manuscript, truncate k_H at radius $C_d\sqrt{\ell}$ and renormalize in L^2 to obtain a smooth radial kernel k_ζ. Its range is at most $C_d\sqrt{\ell}$, and its difference from k_H in L^2, together with derivatives through order three, is at most ζ^{10}. Coupling by the same white noise, let G be the H-layer field for k_ζ and $Z = G - G_H$. The estimates in [B, Proposition 5.5, Step 3] give on a bounded rectangle

$$\sup_x \operatorname{Var} Z(x) \leq C_d\zeta^{20}H, \quad \sup_x \operatorname{Var} \nabla Z(x) \leq C_d\zeta^{20}2^{2H}, \quad \mathbb{E}\sup|Z| \leq C_d\zeta^{10}H.$$

In the present parameters, $\zeta\mathbb{E}\sup|Z| \leq C_d\zeta^{11}H = o(Rg)$. The Gaussian tail at $\zeta\sup|Z| = cRg$ has exponent of order at least $(Rg)^2/(\zeta^{22}H) \gg \ell^2$. Comparing lengths by the factors $e^{\pm\zeta\sup|Z|}$ and initially increasing Q transfers (5.11) to G.

For completeness, the passage to all paths in a unit ball uses directions of mesh ζ^2, and in each direction planes with spacings in $\zeta^2\mathbb{Z}$. Between each pair of planes at distance in $[1/2, 1]$, use transverse sides of length ten centered about the fixed ball center. There are at most $C_d\zeta^{-2(d+1)}$ relevant rectangles. For a path from x to y with $|y - x| = 1$, choose a net direction e with $(y - x) \cdot e \geq 1 - \zeta^4$, and planes between the endpoints at distance at least $1 - 3\zeta^2$. The subarc between its last visit to the first plane before its first visit to the second crosses the associated rectangle. A union bound in (5.11), and $\zeta^2 = o(Rg)$, yield the following event: for every $x \in \overline{B}_{1/4}(z)$, every path from x to $\partial B_1(x)$ in $\overline{B}_1(x)$ has G-weighted length at least

$$1 - (k_d + C(d, Q)R)g. \tag{5.12}$$

Its failure probability is at most $e^{-Q\ell^2}$. The event depends only on noise at radii in $[2^{-H}, 1]$ within distance $C_d\sqrt{\ell}$ of $B_2(z)$. This is exactly the conclusion of [B, Proposition 5.5], now established with (5.1).

5.3 The cover and iteration

We state the cover requirements to check that the longer block is admissible. In [B, Section 6], the cover uses

$$K = C_0\sqrt{\ell}, \qquad a_1 = \lceil K^4 \rceil.$$

For an integer $H \geq 1$ and $\delta \in (0, 1)$ its local events, rescaled to balls of radius $r = 2^{-n}$, must depend on noise at radii $[2^{-n-H}, 2^{-n}]$ within distance $C_d\sqrt{\ell}\, r$ of the ball, and have failure probability at most $e^{-C_* a_1}$. Its remaining conditions are

$$\delta \geq C_* C_0 \zeta \sqrt{a_1}, \qquad \delta \geq e^{-K^2}. \tag{5.13}$$

Here C_0, C_* are the fixed dimensional constants in that proposition. Its conclusion supplies an exceptional set outside which the local crossing estimates hold, the coarse field oscillates by $O_d(\delta)$ after multiplication by ζ, and paths admit detours costing a factor $1 + o(\delta)$. It holds at all $n \geq N_\zeta$, where $\mathbb{P}[N_\zeta > n] \leq C(d, \zeta)e^{-c(d,\zeta)n}$.

Set $\delta = Rg$. Choose $Q > C_*(C_0^4 + 1)$ in Section 5.2. This verifies the required local failure bound, since $a_1 \leq (C_0^4 + 1)\ell^2$ for small ζ. By (5.2), taking B sufficiently large verifies the first inequality in (5.13). The second holds since $e^{-K^2} = \zeta^{C_0^2}$ and C_0 is chosen large. Locality was checked above. These are all the inputs (L1)–(L3) of [B, Proposition 6.1]; in particular its statement allows the present integer H.

We include the exponent extraction. Write h_n for the k_ζ field of the first n dyadic layers, and $D_n(A, B)$ for its internal distance between closed sets $A, B \subset \Omega$. The argument of [B, Theorem 7.2, Steps 1–2] uses the local bound (5.12), the $O_d(\delta)$ oscillation of the coarse field, and the detours to give, for all $n \geq N_\zeta$,

$$D_{n+H}(A, B) \geq \Lambda D_n(A_n, B_n), \qquad \Lambda = 1 - (k_d + C_d R)g. \tag{5.14}$$

Here A_n, B_n are the closed neighborhoods of A, B in Ω of radius $(C_d + 1)2^{-n/6}$. The comparison holds simultaneously for all closed A, B, after increasing the constant in Λ. The endpoint neighborhoods allow the cover's detours to move endpoints.

Fix ζ before iterating. Choose n_0 so that $\mathbb{P}[N_\zeta > n_0] \leq 1/8$ and

$$\rho_0 := \sum_{j \geq 0} (C_d + 1)2^{-(n_0 + jH)/6} \leq 1/4.$$

Choose a deterministic $M_0 < \infty$ so that $\mathbb{P}[\sup_\Omega |h_{n_0}| > M_0] \le 1/8$. On an event of probability at least $3/4$, iterate (5.14) from cutoff $n_0 + JH$ down to n_0. The enlarged endpoint sets stay within $\overline{B}_{\rho_0}(0)$ and $\overline{B}_{\rho_0}(e_1)$. For every integer $J \ge 1$,

$$D_{n_0+JH}(0, e_1) \ge \Lambda^J D_{n_0}(\overline{B}_{\rho_0}(0), \overline{B}_{\rho_0}(e_1)) \ge \tfrac{1}{2} e^{-\zeta M_0} \Lambda^J.$$

Fixed-parameter existence of the distance exponent, or its equivalent median characterization, now implies

$$\lambda_d(\zeta) \le -\frac{\log \Lambda}{H \log 2} \le \{k_d + C_d(R+g)\}\zeta^{4/3}. \tag{5.15}$$

The coupling in [B, Section 2] identifies this exponent with the one for (1.1) at each fixed ζ. Finally $g/R = O(\zeta^{1/3}) = o(1)$. Substituting (5.1) into (5.15) proves

$$\lambda_d(\zeta) \le k_d \zeta^{4/3} + C_d \zeta^{5/3} \ell^{1/2}.$$

Together with (3.1), this completes the proof of Theorem 1.1.

Remark 5.1. All estimates used with growing block sizes are uniform in those block parameters. In particular, Proposition 2.3 is an inequality for every admissible N, n, T, η, and the local crossing estimate was reproved for (5.1). The cutoff limit in the final iteration is taken at fixed ζ and fixed $H = H(\zeta)$.

References

[A] *A small-parameter limit for higher-dimensional Liouville first passage percolation.* Audited manuscript, October 2026, 26 pages. Published PDF.

[B] *Small disorder in log-correlated first passage percolation.* Supplied manuscript, October 2026, 27 pages. File: `lfpp-small-parameter.pdf`.